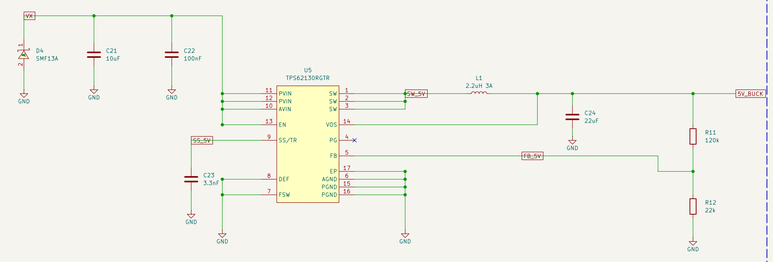
NEXUS-STYLE FLIGHT CONTROLLER | CIRCUIT SCHEMATIC

POWER / COMPUTE + SENSORS / INTERFACES. Named boundaries connect between functional circuits.

UNFINISHED PCB CHECKPOINT — NOT RELEASED FOR FABRICATION. X = intentionally unconnected pin.

01 VX INPUT / 5 V BUCK

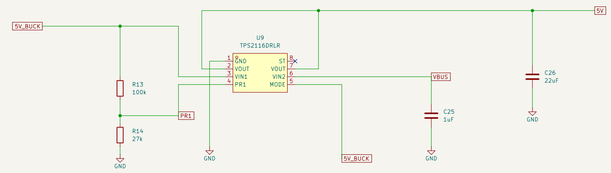
TPS62130: 3 A, 2.5 MHz | Vout = 0.8 V x (1 + 120k / 22k) = 5.16 V



FSW = GND: 2.5 MHz; DEF = GND

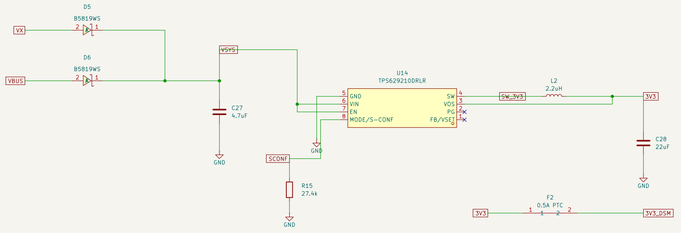
02 5 V SOURCE PRIORITY

Buck-priority power mux | 5V_BUCK preferred, USB VBUS fallback



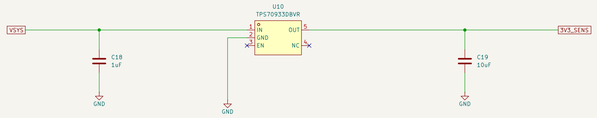
03 VSYS / MCU 3.3 V

Diode-OR supply -> TPS629210 -> MCU rail; fused DSM supply branch



04 LOW-NOISE SENSOR SUPPLY

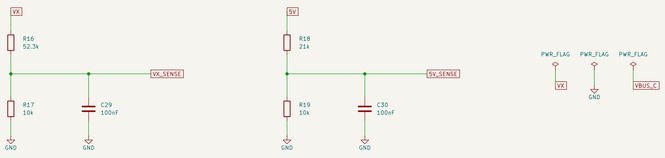
TPS70933: EN pin 3 intentionally floating, supported by internal pull-up



EN and NC are intentionally unconnected. Do not tie EN to VSYS.

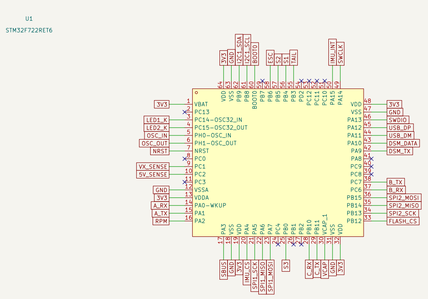
05 VOLTAGE MONITORING / POWER FLAGS

ADC divider filters | VX_SENSE -> U1.9; 5V_SENSE -> U1.10



06 STM32F722 CORE

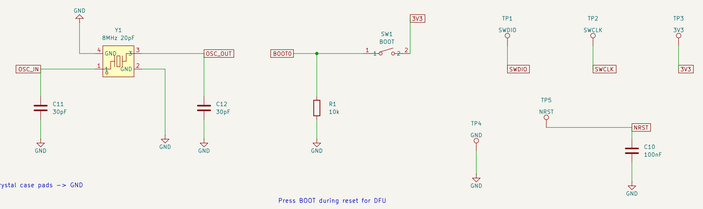
Rotorflight RDMS-NEXUS_F7 pin map: boundary labels connect the functional units



SPI1: inertial sensor | SPI2: blackbox | I2C1: barometer

07 CLOCK / BOOT / DEBUG

8 MHz HSE with load capacitors; BOOT0 button and pull-down; SWD / reset pads

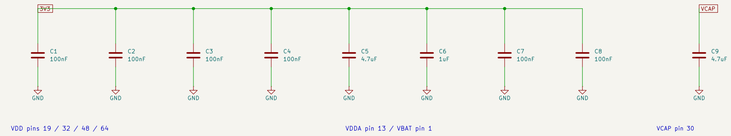


Crystal case pads -> GND

Press BOOT during reset for DFU

08 MCU DECOUPLING

Local 3V3 bypass: C1–4 VDD, C5 bulk, C6–7 VDDA, C8 VBAT; C9 VCAP only



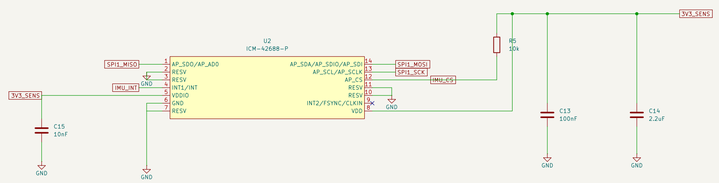
VDD pins 19 / 32 / 48 / 64

VDDA pin 13 / VBAT pin 1

VCAP pin 30

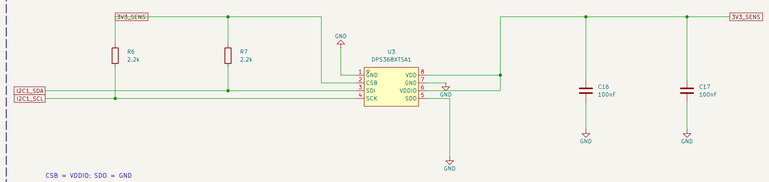
09 SPI1 INERTIAL SENSOR

ICM-42688-P: IMU_CS pull-up maintains SPI mode at startup; IMU_INT -> U1.50



10 I2C1 PRESSURE SENSOR

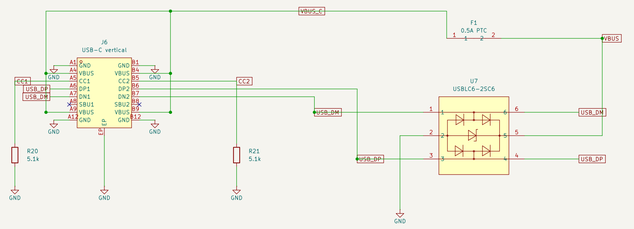
DPS368 address 0x76: SDO low; CSB high selects I2C; local SCL / SDA pull-ups



CSB = VDDIO; SDO = GND

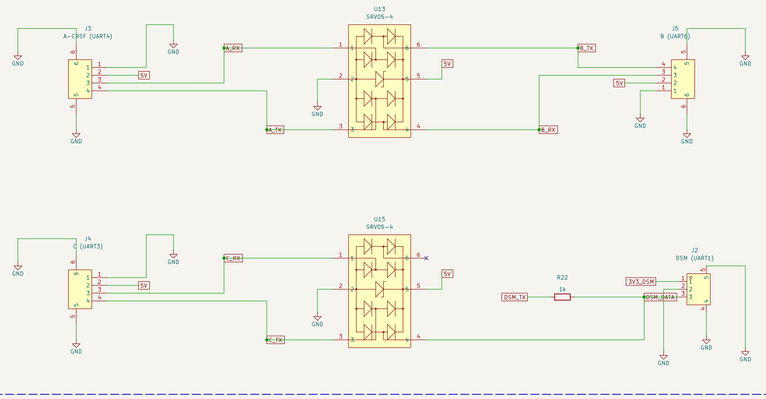
11 USB-C / PROTECTION

USB 2.0 device | CC pull-downs: 5.1k | F1: VBUS resettable fuse



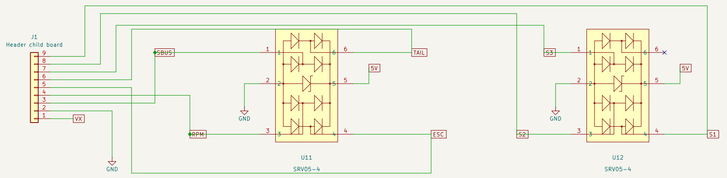
12 RECEIVER INTERFACES

A: UART4 / CRSF | B: UART6 | C: UART3 | DSM: UART1



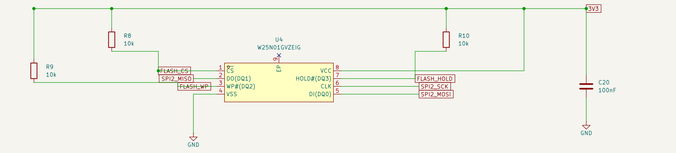
13 SERVO HEADER / SIGNAL CLAMPS

J1 connects the separate header board: VX / GND enter at pins 1 / 2



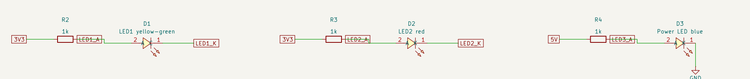
14 SPI2 BLACKBOX

W25N01GV 128 MB | /CS, /WP, /HOLD pull-ups | EP intentionally floating



15 STATUS INDICATION

PC14 / PC15 sink current: active low | Blue indicates 5 V bus power



Rotorflight RDMS-NEXUS_F7 target; LCSC part numbers in the LCSC field
Generated from tools/design.py — edit the design file, then regenerate

Sheet: /
File: nexus-style-fc.kicad_sch

Title: Nexus-style FBL controller — main board

Size: A1	Date: 2026-10-13	Rev: 0.2
KiCad E.D.A. 10.0.7		Id: 1/1